

Module 6: Lesson 2

Multi-Armed Bandits: Stationary Case



Outline

- ▶ Stationary k -Armed Bandit problem: Actions and learning.
- ▶ Results from a k -Armed Bandit testbed.

A stationary k -Armed Bandit problem: Actions and learning

We are going to focus Lesson 2 on the implementation of optimal choices in the context of a stationary k -armed bandit problem.

The core of the algorithm is made of two pieces within each time step. First, an optimal action choice, given information about the past.

$$A_t = \arg \max_a Q_t(a) \quad (1)$$

We are going to compare the performance of this greedy policy with an ε -greedy policy that chooses a random action with probability $\varepsilon = 0.1$.

Second, an updating of the expected reward from choosing the given action.

$$Q_t(a) = Q_{t-1}(a) + \frac{1}{N_t(a)}[r_t - Q_{t-1}(a)] \quad (2)$$

A stationary k -Armed Bandit problem: The setup

We will study the learning ability of an agent in the context of a 10-armed bandit problem.

The average payoff from each bandit comes from the draws of a standard normal distribution:

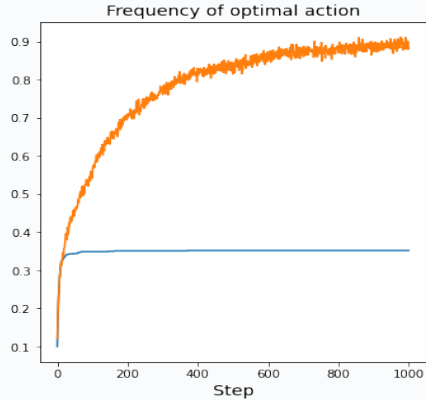
$$[0.47, -1.19, 1.43, -0.31, -0.72, 0.89, 0.86, -0.64, 0.04, -2.24] \quad (3)$$

The rewards of choosing each arm come from a normal distribution centered at the corresponding mean, with a unit standard deviation.

We are going to evaluate the performance of the agent to learn over a run, or “episode”, of 1,000 time steps.

We perform the exercise 1,000 times, more than enough to reduce noise. We check the learning ability of the agent by averaging, at each time step, the rewards obtained by the agent across all episodes.

A stationary k -Armed Bandit problem: Results



A stationary k -Armed Bandit problem: Results

The greedy method learns slightly faster than the ε -greedy method at the beginning, but then reaches a lower average reward.

The greedy method finds the optimal action in roughly one-third of the steps.

The greedy method performs worse in the long run because it gets stuck performing suboptimal actions.

The ε -greedy method eventually performs better because of its exploration of other actions in search of the optimal one.

The long-term performance of a low ε is better: Reduce the parameter over the steps to maximize the advantages of high and low values.

The advantage of ε -greedy over greedy methods depends on the task: high vs. low variance of rewards across actions.

Summary of Lesson 2

In Lesson 2 we have looked at:

- ▶ Understand the workings of a k -Bandit problem in a stationary setup.
- ▶ Understand the advantages and disadvantages of the ϵ -greedy actions.

⇒ **References for this Lesson:**

Sutton, R. S., Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press. (Chapter 2)

TO-DO NEXT: Now, please go to the associated Jupyter Notebook of this Lesson to understand the coding implementation of the stationary k -Armed Bandit problem.

In the Next Lesson we will cover the non-stationary case of the k -Armed Bandit problem.